MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-M101/BSM101 Mathematics –IA

UPID : 001004

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) f(x)=|x| satisfies Roll's theorem [-1,1]. State true or false with reasons.

(ii) Every square matrix A can be expressed as a sum of symmetric and skew symmetric matrix and it can be expressed as=

(iii) Collection of vectors containing null vectors is linearly _____.

(iv) The eigen value of the matrix A are a and b; then what are the eigen values of A^2 ?

(v) The value of $\Gamma(5/2) = ?$

(vi) $\lim_{x\to 0} \frac{1-Cos\,x}{Sin\,x} = ?$

(vii) Find the value of C for which the vector (1,2) and (3,C) linearly independent.

(viii) A set of vectors in a vector space V over a field F is orthogonal if $||\alpha|| =$ _____.

(ix) What is eigen value of a square matrix? orthonormal.

Find the eigen value of the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

(x) Define Beta and Gamma functions.

(xi) The product of two Orthogonal matrix is _____ matrix.

(xii) The product of any matrix and its transpose is _____ matrix.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Check whether the vectors { (1, 1, 0), (1, 1, 1), (0, 1, -1) is a basis in R^3 or not. [5]

3. State Lagrange's mean value theorem. [5]

Prove that $\frac{x}{1+x} < \log(1+x) < x \;, x > 0$

4. Solve the following equations by matrix inversion method, if possible: [5]
2x+3y+z= 6; 3y+5z= 8; -x+y=0

5. Define subspace of a vector space. [5]

Let V=$\{M_2$ *set of all* 2×2 *matrices of the form* $\begin{pmatrix} a & b \\ c & d \end{pmatrix}, a,b,c,d \in R\}$ and S is the ~~subset~~ subspace of V such that $S=\{\begin{pmatrix} a & b \\ c & d \end{pmatrix} : a+b=0\}$ is a subspace of V.

6. Prove that B(m, n) = B(n, m). Also write the relation between Beta and Gamma functions and find B(3,4). [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Define kernel of a linear transformation between two vector spaces. [5]
Is $f(x) = Sin\,x$ a Linear transformation? Justify your answer.

(b) State Cayley Hamilton theorem for matrices. What is the Characteristic equation of a square matrix? Explain with an example. [4]

(c) What is eigen vector? [6]

Find the eigen values and eigen vectors of matrix $\begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}$.

8. Show that the set of all complex numbers is a vector space over real field w.r.t the usual addition between two complex number and multiplication between a real number and a complex number. Find a basis of this space and hence the dimension. [15]

9. What is diagonalization of matrix? [15]

Diagonalise the matrix $A = \begin{pmatrix} 6 & 4 & -2 \\ 4 & 12 & -4 \\ -2 & -4 & 13 \end{pmatrix}$. Write the matrix P, such that $P^{-1}AP$ is diagonal matrix. Also show that $P^{-1}AP = D$.

10. (a) Prove that $\int_0^\infty \frac{1}{(1+x^2)^5} dx = 35\pi/256$ [7]

(b) Obtain the evolute of the parabola $y^2 = 4ax$. [8]

11. (a) Verify Lagrange's MVT for $f(x) = x(x-1)(x-2)$ in $[0, 1/2]$. [5]

(b) Expand the following functions in power of x in infinite series stating in each case the condition under which the expression is valid: $f(x) = \cos x$. [5]

(c) Evaluate $\Gamma(1/2)\ \Gamma(3/2)\ \Gamma(5/2)\ \Gamma(7/2)$. [5]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : ES-EE101/ESEE101 Basic Electrical Engineering

UPID : 001009

Time Allotted : 3 Hours *Full Marks : 70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) What is the Power factor of electric incandescent bulb?

(ii) Does the transformer draw any current when secondary is open? Why?

(iii) Explain the law based on which a DC generator works.

(iv) Calculate peak-peak voltage if V_{max}=1 V and V_{min}=-1 V.

(v) What is the full form of MCCB?

(vi) Draw the Norton's equivalent circuit

(vii) What is the relation between line voltage and phase voltage in star connection?

(viii) Why are iron losses considered as constant losses in transformer?

(ix) Why is the starting current high in a DC motor?

(x) A single-phase full bridge inverter has a dc voltage source Vs = 230 V. Find the rms value of the fundamental component of output voltage.

(xi) In an AC circuit, how many types of power can be defined?

(xii) If we place a short circuit link across a resistance then the resistance doesn't have any effect of the circuit. Explain

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Explain the working principle of single-phase transformer. [5]

3. A. d.c. Shunt generator supplies a load of 50KW at 500 V. Calculate the emf generated if the armature and shunt field resistances are 0.25 ohm and 125 ohm respectively. [5]

4. Explain the operation of Buck converter. [5]

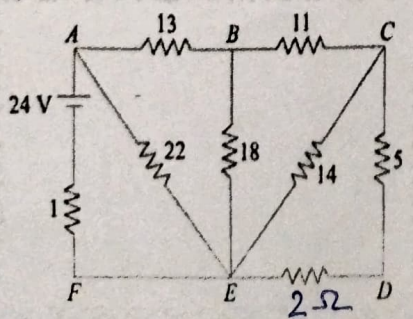
5. A 60-KVA single phase transformer with a primary voltage of 2,400 volts and a secondary voltage of 240 volts. a. What is the rated current in the primary b. what is the rated current in the secondary. c. find the turns ratio. [5]

6. Three parallel connected resistors when connected across a d.c. voltage source dissipate a total power of 72 W. The total current drawn is 6 A, the current flowing through the first resistor is 3 A and the second and third resistors have equal value. What are the resistances of the three resistors? [5]

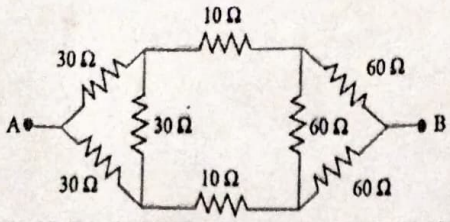
Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Using series and parallel combinations for the electrical network of Fig, calculate (a) current flowing in branch AF (b) p.d. across branch CD. All resistances are in ohms. [7]



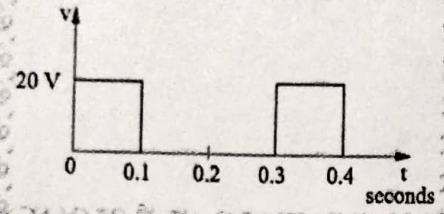
(b) Find the resistance between the terminals A and B for the network shown in Fig. [6]



(c) Explain linear and bilateral network. [2]

8. (a) Define r.m.s. and average value as applied to ac voltage, prove that in pure inductive circuit, current lags behind applied voltage at an angle 90°. [8]

(b) Compute the average and effective values of the square voltage wave shown in Fig. [7]



9. (a) Derive the EMF equation of a transformer. [4]

(b) A 8 kVA, 440/2000 V, 50 Hz single phase transformer gave the following test results: [8]
No load test: 400 V, 0.8 A, 80 W
Short circuit test: 50 V, 3 A, 20W
Calculate (i) the magnetizing current and the component corresponding to iron losses at normal voltage and frequency, (ii) the efficiency on full load at unity power factor, (iii) the secondary terminal voltage on full load at unity power factor.

(c) Draw the equivalent circuit of a transformer refer to the primary side. [3]

10. (a) Derive the emf equation of dc machine. [4]

(b) A dc motor has an armature resistance of 0.9 Ω and takes an armature current of 18 A from 230 V dc mains. Calculate the power output and overall efficiency of the motor if the rotational losses are measured to be 112 W and the shunt field resistance is 300 Ω. [7]

(c) Write the advantages of a three phase induction motor [4]

11. (a) What is the r.m.s value of an a.c. quantity? Obtain expression for the r.m.s. value of a sinusoidal current in terms of its maximum value. [5]

(b) A balanced star-connected load of (8 + j6) Ω per phase is connected to a balanced 3-phase 400-V supply. Find the line current, power factor, power and total volt-amperes. [4]

(c) Three resistances of 5, 10 and 15 Ω are connected in delta across a 3-phase supply. Find the values of the three resistors, which if connected in star across the same supply, would take the same line currents. [6]

*** END OF PAPER ***

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-PH101/BSPH101 Physics-I

UPID : 001003

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) Write an expression for Compton Wavelength of an electron.

(ii) MB statistics is applicable for photons. State True or False.

(iii) What is the displacement of a particle executing SHM, when the velocity is maximum?

(iv) What is poulation inversion?

(v) For which type of materials magnetic susceptibility is negative?

(vi) Phase velocity and group velocity are equal when the medium is _____ .

(vii) Which statistical distribution is followed by free electrons in a conductor?

(viii) Write the differential equation for SHM.

(ix) What is Optic Axis of a crystal?

(x) What is the dimension of $\mu_0\epsilon_0$?

(xi) An electron and a proton are accelerated through the same potential difference. The de Broglie wavelength of the proton is larger than that of the electron. State True or False.

(xii) Give one example of Uniaxial Crystal.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. a) Draw the plot for blackbody radiation against wavelength for three different temperatures T1<T2<T3 [5]
b) What do you mean by Ultra-Violet Catastrophe?

3. (a) Show that the gradient of a scalar field u(x,y,z) is perpendicular to the surface u(x,y,z)=constant [5]
(b) Find Gradient of the following scalar field at the point (1, -2, -1).
$u(x,y,z) = xy^2z + 4x$

4. State Brewster's Law. Calculate the polarizing angle for light traveling from water to glass. Refractive index of glass=1.53 and water =1.33 [5]

5. Differentiate between Amplitude resonance and velocity resonance. What is logarithmic decrement in damped oscillation? [5]

6. A particle is trapped in an infinite one dimensional potential well of width 'L' . Write its wavefunction and probability density function in its first excited state. Find the probability density at x=L/2. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Show that the gradient of a scalar function is an irrotational vector, [3]

(b) Show that Curl of position vector is zero. [3]

(c) Is the magnetic field an irrotational vector or a solenoidal vector? Is the electric field an irrotational vector or a solenoidal vector? [2]

(d) Solve the following . [4]
If $\psi(x,y,z) = 3xy^2 - 5x^2z + 2z^2$, find $\nabla^2\psi$

(e) Determine the constant 'q' so that the following field is solenoidal [3]
$\vec{F} = (2x + 3yz)\hat{i} + (3y - z^2)\hat{j} + (y^3 + qz)\hat{k}$

8. (a) Distinguish between Fresnel and Fraunhofer Diffraction [3]

(b) Write the characteristics of LASER. [4]

(c) Mention three methods of producing plane polarized beam of light. [3]

(d) Explain briefly what do you mean by double refraction. [3]

(e) State Malus law. [2]

9. (a) What is Grating element? Write and plot the intensity distribution of a single slit diffraction pattern and find the positions of maxima, minima and secondary maxima. [2+1+2+4]

(b) Find the minimum thickness of a half wave plate and a quarter wave plate for a light wave of wavelength 5893 Angstrom. Given $\mu_0 = 1.66$ and $\mu_e = 1.47$ [2]

(c) What are the three basic components of a LASER? What is a optical resonator? [2+2]

10. (a) X-Ray photon of wavelength 2 angstroms is scattered at an angle of 45 degrees by a carbon block. [2+3]
(i) Calculate the wavelength of scattered X ray
(ii) Find the fraction of energy lost by the X-Ray photon during this event.

(b) A particle of mass m is confined in an infinite potential well between x=0 and x=L. Solve the time independent Schrodinger equation for such a system and Find the normalized eigenfunctions and eigenvalues. Using the normalized eigenfunctions find the expectation value of x. [5+2]

(c) Plot the Fermi distribution of a particle at T=0 K and T>0 K. [3]

11. (a) Calculate the number of photons emitted per sec by a 20 W source of monochromatic light of wavelength 431 nm. [4]

(b) What do you mean by degeneracy? [2]

(c) State the Heisenberg's Uncertainty Principle. The average time interval that elapses between the excitation of an atom and the time it radiates energy is 10^(-8) seconds. Calculate the uncertainty in energy of emitted photons. [2+3]

(d) Compare Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac Statistics on the basis of minimum three characteristics. [4]

*** END OF PAPER ***